

Formula Sheet for Chapter 9

The Discounted Dividend Model

$$\hat{P}_0 = \frac{D_1}{(1+r_s)^1} + \frac{D_2}{(1+r_s)^2} + \frac{D_3}{(1+r_s)^3} + \dots + \frac{D_\infty}{(1+r_s)^\infty}$$

- $D_1, D_2, \dots$ = expected future dividends
- P_0 = current stock price
- r_s = required rate of return (based on risk)

The Constant Growth (Gordon) Model

$$\hat{P}_0 = \frac{D_0(1+g)^1}{r_s - g} = \frac{D_1}{r_s - g}$$

- g = expected growth rate of dividends

Expected Return from (Gordon) Model

$$\hat{r}_s = \frac{D_1}{P_0} + g$$

- Dividend Yield = D_1 / P_0
- Capital Gains Yield = % change in price = g
- Total Return = Dividend Yield + Capital Gains Yield

$$g = (1 - \text{Payout Ratio}) \times ROE$$

$$FCF = EBIT(1-T) + \text{Dep \& Amort} - (\text{Capital Expenditures} + \Delta \text{ Net Operating Working Capital})$$

The Corporate Valuation Model:

Market value of company's operations =

$$\frac{FCF_1}{(1+WACC)^1} + \frac{FCF_2}{(1+WACC)^2} + \frac{FCF_3}{(1+WACC)^3} + \dots + \frac{FCF_\infty}{(1+WACC)^\infty};$$

$$\text{Horizon value}_0 = \frac{FCF_0(1+g)^1}{WACC - g} = \frac{FCF_1}{WACC - g};$$

Market Value of Company = MV of company's operations + MV of company's non-operating assets;

$$\text{Intrinsic Value per Share} = \frac{\text{Market Value of Company} - \text{debt} - \text{preferred stock}}{\text{\#shares}}$$

$$\text{Value of Preferred Stock} = \frac{D_p}{r_p};$$

- D_p : Annual Dividend
- r_p : Required Return

